Kennesaw State University 2026-2027 Undergraduate Catalog

Courses

Contract All Courses |

Information Technology

- ### IT 1103: Introduction to Information Technology

3 Credit Hours

Concurrent: IT 1114

Description

This course introduces information technology (IT) as a major computing discipline and profession. The course covers fundamental IT concepts, processes, systems, and basic components including computer system, software, networking, cybersecurity, and data. Students will also have a general knowledge of IT profession and career.

Course Learning Outcomes

Students who successfully complete this course will be able to:

1. Compare IT to related computing fields/disciplines and other related disciplines.
2. Explain major components of IT infrastructure and major knowledge areas.
3. Explain major IT job roles, types, and career paths.

- ### IT 1113: Introduction to Programming

3 Credit Hours

Description

This course covers the fundamentals of computer programming. Concepts of counters, accumulators, decision-making, looping, subroutines, arrays, files, and string processing are covered. A programing language is used for assignments.

- ## IT 1114: Programming Principles

3 Credit Hours

Concurrent: IT 1114L

Description

This course introduces the principles of computing and programming. Upon completion of this course, students will apply the basic concepts of structured programming through design, coding, debugging, and documenting. This will include basic syntax and semantics for sequential, conditional, and iterative control structures, design & use of basic data structures and functions.

Course Learning Outcomes

Students who successfully complete this course will be able to:

1. Describe the fundamentals of data representations and how those apply to modern computing.
2. Describe the fundamentals of algorithm selection and complexity.
3. Explain basic programming principles such as logical expressions, selection, and repetition.
4. Design and modularize solutions with proper use of functions and objects.
5. Solve programming problems which include searching and sorting of arrays.
6. Describe how style guidelines enhance readability and promote correctness in programs.

- ## IT 1114L: Programming Principles Lab

1 Credit Hours

Concurrent: IT 1114

Description

This course provides lab activities to accompany IT 1114. Upon completion of this course, the student will design, code, debug, document and apply the basic concepts of structured programming. This will include basic syntax and semantics for sequence, conditional, and iteration control structures, (design & use of functions) as well as single dimensional arrays. The student will be able to solve problems by designing and modularizing their solutions with proper use of functions and usage of objects.

- ## IT 2290: Special Topics

1-4 Credit Hours

Prerequisite: IT 1113 & permission from Department

Description

The course covers special topics at the intermediate level that are not in the regular course offerings.

- ## IT 3123: Hardware and Software Concepts

3 Credit Hours

Prerequisite: ((IT 1114 and IT 1114L) or (IT 1113) or (CSE 1321 and CSE 1321L))

Description

This course examines various hardware and software components and how they work together in a modern computing environment. Topics include an overview of computer organization and architecture, machine language and modern languages.

Course Learning Outcomes

Students who successfully complete this course will be able to:

1. Identify, use, and convert binary, decimal, and hexadecimal number systems, describe common digital data formats, and describe fundamental digital logic circuits and their relationship to binary numbers.
2. Explain the relationship between digital logic and computation, describe the parts of a computer instruction, and explain the fetch-decode-execute instruction cycle.
3. Describe and use the components of a CPU, input/output hardware, peripherals, and digital communications and how they work. Apply the hierarchy of storage to explain how storage, data, and computation are related.
4. Describe the functions of an operating system and be able to compare various types of operating systems, including file management, cache and virtual memory, memory management, and security.
5. Use and differentiate between high-level, low-level and machine computer languages.

- ## IT 3203: Introduction to Web Development

3 Credit Hours

Prerequisite: (IT 1114 and IT 1114L with 'C' or better) OR (IT 1113 with 'C' or better) OR (CSE 1321 and CSE 1321L with 'C' or better)

Description

This introduction course covers applications for the world wide web. Topics include current languages (such as HTML and JavaScript), basic web protocols, and human-computer interfaces for the web.

- ## IT 3223: IT Organization and Management

3 Credit Hours

Description

This course provides a high-level introduction to several areas that are crucial to the Information Technology and Security profession: IT management, hardware/software acquisition, software development processes, and project management. This course will provide students with a foundation in understanding of planning, control, operating, and decision-making processes including project scope, request for proposals, and the software development life cycle. The course will introduce students to the formal organization of IT and a widely used project management information system.

Course Learning Outcomes

Students who successfully complete this course will be able to:

1. Discuss the elements of the basic Software Development Life Cycle and implement an analysis and design of a project using current project management tools.
2. Compare and contrast the advantages and disadvantages of building vs buying a software system and evaluate a Request for Proposal for a software package.
3. Describe the project phases and challenges of acquiring, integrating, and implementing a software package.
4. Identify the elements and challenges of projecting scoping and project management.
5. Demonstrate the ability to use Project Management tools and technology available to define, plan and monitor projects
6. Identify the basic tenants areas of managing Information technology and information systems in organizations, management, systems, and services.

- ## IT 3423: Operating Systems Concepts & Administration

3 Credit Hours

Prerequisite: IT 3123 or CS 3503.

Description

This course is an introduction to basic operating system principles. Topics include memory management, peripheral device management, file system management and process management. Different types of operating systems and their administrations are studied. Projects are carried out with simulations.

- ## IT 3503: Foundations of Health Information Technology

3 Credit Hours

Prerequisite: ENGL 1102

Description

This course introduces students to the field of health information technology (HIT). Students will become familiar with the content, use, and structure of the health care data and medical records, health information management, the fundamentals of healthcare workflow and process analysis and redesign as a necessary component of complete practice automation, and health information systems. Students will also become familiar with the health care delivery systems in the U.S. and IT organizations and resources.

- ## IT 3703: Introduction to Data Analytics and Technology

3 Credit Hours

Prerequisite: IT 3123
Concurrent: CSE 3153 or CS 3410
Description
This course provides a comprehensive overview of processing technologies, systems, and application. Latest trends in data analytics, big data, and data science will be introduced.

- ## IT 3883: Advanced Application Development

4 Credit Hours

Prerequisite: ((IT 1114 and IT 1114L, minimum grade of 'C') or (CSE 1321 and CSE 1321L, minimum grade of 'C')) and (CSE 3153 or CS 3410, minimum grade of 'C')
Description
This course will allow students to learn the advanced aspects in application development, including graphical user interface, business logic, algorithm analysis, and data/content processing. Students will design and evaluate effective user experience for IT applications and systems. This course will include hands-on exercises for experience with processes and tools.

Course Learning Outcomes
Students who successfully complete this course will be able to:

1. Analyze real-world problems and design algorithmic/programming solutions.
2. Develop requirements and evaluation protocols for effective user experience design within IT systems including those for accessibility.
3. Write, manage, analyze, test, and debug program code in accordance with modern software engineering principles.
4. Apply basic software engineering guidelines for gathering requirements, designing, and building an application.

- ## IT 4153: Advanced Database

3 Credit Hours

Prerequisite: CSE 3153 or CS 3410
Description
This course will study how databases are used with programming applications. Topics include advanced PL/SQL (or similar database programming language), database transaction, database security, database maintenance, and distributed and web databases.

- ## IT 4323: Data Communications & Networking

3 Credit Hours

Prerequisite: IT 3123 or (CS 3503)

Description

Fundamental concepts of computer networking include topics such as properties of signals and media, information encoding, error detection and recovery, LANs, backbones, WANs, network topologies, routing, Internet protocols, and security issues. The focus is on general concepts together with their application to support the business enterprise.

- ## IT 4333: Network Configuration & Administration

3 Credit Hours

Prerequisite: IT 4323 or ECET 3400 or CS 4622

Description

This course continues the study of networks. Topics include design and implementation of networks including synchronization, scheduling, exception and deadlock resolution, client server and web based collaborative systems. Network security will also be covered. Cost estimates and speed are examined from a management perspective.

- ## IT 4400: Directed Studies

1-3 Credit Hours

Prerequisite: Approval of the instructor and department chair

Description

This course covers special topics of an advanced nature that are not in the regular course offerings. Students will complete a research project in the computing field supervised by a faculty member. Credit hours vary from one to three depending on the nature and content of the project student involved. Up to three credits may be applied to the major area.

- ## IT 4403: Advanced Web and Mobile Applications

3 Credit Hours

Prerequisite: IT 3203

Description

This course covers advanced topics on web and mobile applications with a focus on modern web application architectures and mobile friendly user interfaces. Students will complete one major development project using the latest web technologies.

- ## IT 4423: Linux/Unix Administration

3 Credit Hours

Prerequisite: IT 3423 or CS 3502

Description

This course introduces Linux/Unix operating systems. Topics include system administration, file systems and access permissions, regular expression, common tools and utilities, and network service configurations. Lessons will be enhanced using hands-on exercises.

- ## IT 4490: Special Topics in Information Technology

1-3 hours Credit Hours

Prerequisite: Vary by specific topic, Junior/Senior Standing

Description

Special topics selected by the Department Chair. Offered on a demand basis.

- ## IT 4493: IT Undergraduate Research

3 Credit Hours

Prerequisite: 60 Credit Hour Completion (junior standing)

Description

This course promotes undergraduate research in information technology. Students develop research ideas and conduct research to investigate topics with the guidance of an IT faculty member. Students document their finding in final reports, present their findings, and prepare research papers for publication in appropriate venues.

- ## IT 4513: Electronic Health Record Systems & Applications

3 Credit Hours

Prerequisite: IT 3503

Description

This course provides an overview of the importance of key technical aspects of electronic health records, the overall architecture, features and functions of major EHR systems. Hands-on exercises with EHR systems allow students to learn by doing.

- ## IT 4523: Clinical Processes and Workflows: Analysis and Redesign

3 Credit Hours

Prerequisite: IT 3503 or concurrent

Description

This course introduces the fundamentals of healthcare workflow and process analysis and redesign as a necessary component of complete practice automation. Students will become familiar with the concepts of processes, process analysis and redesign in the healthcare settings. Workflow and process mapping in healthcare improvement including detailed guidance, helpful tools, and case studies are introduced.

- ## IT 4533: Health information Security and Privacy

3 Credit Hours

Prerequisite: IT 3503

Description

Health information security and privacy are of utmost importance in today's healthcare environment. This course introduces the concepts, practices and concerns of information privacy and security unique to the healthcare settings. The course provides the student with a thorough understanding of the HIPAA security and privacy rules, meaningful use security requirements, security risk assessment and management, and how to integrate privacy and security into medical practices. Resources of privacy and security in healthcare are provided.

- ## IT 4673: Virtual IT Systems

3 Credit Hours

Prerequisite: (IT 3423 or CS 3502) and IT 4323 and (CSE 3153 or CS 3410)

Description

This course explores the design, implementation and use of virtualization, including desktop and server aspects such as deployment, clustering, storage and security. A high level overview of the various certifications available will be discussed. A project will be completed as part of the course.

- ## IT 4713: Business Intelligence Systems

3 Credit Hours

Prerequisite: IT 3703
Description
This course introduces the concepts, practices, technologies and systems of business intelligence, which supports enterprise level data management, analytical processing, and reporting.

- ## IT 4723: IT Policy and Laws

3 Credit Hours

Prerequisite: IT 3123 and IT 3223
Description
This course covers current policies and law, and ethical, legal, and social issues in IT through lectures, discussion, research, and case studies. Topics include copyright, patents, trademarks, trade secrets, computer ethics, computer crime, computer abuse, cultural impact, web issues, information warfare and current legislation.

- ## IT 4733: Big Data System Administration

3 Credit Hours

Prerequisite: IT 3703
Description
This course introduces contemporary distributed big data platforms and database systems. Topics include big data computing paradigms, big data platform architectures and administration, and big data database concepts and administration.

- ## IT 4773: Machine Learning for Enterprise Applications

3 Credit Hours

Prerequisite: IT 3703
Description
This course covers enterprise applications of machine learning on data analytics. Topics span the complete life cycle of data analytics from envisioning the problem, identifying proper data, selecting suitable machine learning methods, evaluating modeling performance, and deploying the results to improve decision-making.

- ## IT 4793: Applied Data Driven Solutions

3 Credit Hours

Prerequisite: IT 3703

Description

This course will apply data technologies and applications in a chosen application domain and expose students to real world problems and issues. Students will analyze, design, and develop a practical and technical data driven solution in a chosen field, such as healthcare, finance, security, social media, academia, sports, and utilities.

- ## IT 4823: Information Security Administration & Privacy

3 Credit Hours

Prerequisite: (CSE 2300 or MATH 2345) and (IT 3123 or CS 3503)

Description

The student develops knowledge of the principles of information assurance at the policy, procedural, and technical levels to prepare the student for a role as a business decision-maker. Real-world examples from the text and current events will be used to demonstrate the applicability of the techniques of information assurance.

- ## IT 4833: Wireless Security

3 Credit Hours

Prerequisite: IT 4323 or ECET 3400 or CS 4622

Description

This course covers methods and techniques to secure wireless networks against threats and attacks. The topics may include Security and Cryptography, Network Security Protocols, Security and Layered Architecture, Voice-Oriented Wireless Networks, Data-Oriented Wireless Networks, Security in Traditional Wireless Networks, Security in Wireless LAN, Security in Wireless Ad Hoc Network, Special Topics such as Mobile Security.

- ## IT 4843: Ethical Hacking for Effective Defense

3 Credit Hours

Prerequisite: IT 4323 or ECET 3400 or CS 4622

Description

This course focuses on detection of network and system vulnerabilities by taking an attacker-like approach to system, network, and data access. Topics include network attacks and defenses, Operating System and application vulnerabilities, social engineering attacks, and malware. Ethical, legal implications of network attacks are also discussed.

- ## IT 4853: Computer Forensics

3 Credit Hours

Prerequisite: IT 4323 or CS 4622 or ECET 3400

Description

This course studies techniques and tools in computing investigation, digital evidence collection, recovery, and analysis. Topics include: Legal issues relating to digital evidence, recover deleted files and discover hidden information, reconstruct user activity from e-mail, temporary Internet files and cached data, assess the integrity of system memory and process architecture to reveal malicious code.

- ## IT 4863: Web and Mobile Application Security

3 Credit Hours

Prerequisite: IT 3203

Description

This course introduces web and mobile application security issues, hands-on practices to explore security vulnerabilities, and best practices to defend against vulnerabilities in web and mobile.

- ## IT 4883: Infrastructure Defense

3 Credit Hours

Prerequisite: IT 4323 or ECET 3400 or CS 4622

Description

This course provides an overview of the infrastructure assessment and penetration testing process and the processes and techniques for improving the defensibility of that infrastructure.

- ## IT 4893: Internet of Things: Applications and Security

3 Credit Hours

Prerequisite: IT 4823 and IT 4323

Description

This course introduces core knowledge and skills required to develop and design innovative IoT solutions. Students will analyze the challenges, apply appropriate patterns for user-interaction and learn about trends and characteristics in IoT. In addition, students will evaluate the security design of a suite of IoT-connected products.

- ## IT 4983: IT Capstone

3 Credit Hours

Prerequisite: IT 3423 and IT 3223 and IT 3203 and IT 3883

Concurrent: IT 4323 and IT 4823

Description

Students work in teams to research, develop, or implement an IT solution integrating the knowledge acquired in preceding IT courses. Key components emphasized include technical design, research, documentation, project management, leadership, teamwork, and effective communication skills. The final outcome will be an IT solution or result addressing a typical business, organizational, or academic need.

Course Learning Outcomes

Students who successfully complete this course will be able to:

1. Apply IT concepts, applications, best practices, and standards to create a solution or report that addresses real business needs.
2. Practice soft skills such as project planning and management, research and learning, communication, writing, presentation, and team work, in a real world IT project.
3. Produce a career portfolio for use in employment or graduate applications.

Contract All Courses |